

Lab Experiments

Course Code/Course Name: DSA0604/Data Handling and Visualization

Name/ Reg No: G. Sahitya (192424217)

Set 11: Product Category Analysis

R code:

```
library(ggplot2)
library(plotly)
library(tidyr)
```

```
# Dataset
```

```
weather_data <- data.frame(
  Date = as.Date(c("2025-01-01", "2025-01-02", "2025-01-03",
    "2025-01-04", "2025-01-05")),
  Temperature = c(28, 30, 29, 31, 32),
  Humidity = c(65, 70, 68, 72, 75),
  Rainfall = c(5, 10, 0, 8, 12)
)
```

```
# 1. Line Chart
```

```
ggplot(weather_data,
  aes(x = Date, y = Temperature, group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Daily Temperature Trend",
    x = "Date",
    y = "Temperature (°C)"
  ) +
  theme_minimal()
```

```
# 2. Scatter Plot
```

```
ggplot(weather_data,
  aes(x = Humidity, y = Temperature)) +
  geom_point(color = "darkgreen", size = 4) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(
    title = "Humidity vs Temperature",
    x = "Humidity (%)",
    y = "Temperature (°C)"
  ) +
  theme_minimal()
```

3. Area Chart

```
ggplot(weather_data,  
  aes(x = Date, y = Rainfall)) +  
  geom_area(fill = "skyblue", alpha = 0.6) +  
  labs(  
    title = "Daily Rainfall",  
    x = "Date",  
    y = "Rainfall (mm)"  
  ) +  
  theme_minimal()
```

4. Interactive Dashboard

```
p1 <- ggplot(weather_data,  
  aes(x = Date, y = Temperature)) +  
  geom_line(color = "blue") +  
  geom_point(color = "red") +  
  labs(title = "Temperature Trend")
```

```
ggplotly(p1)
```

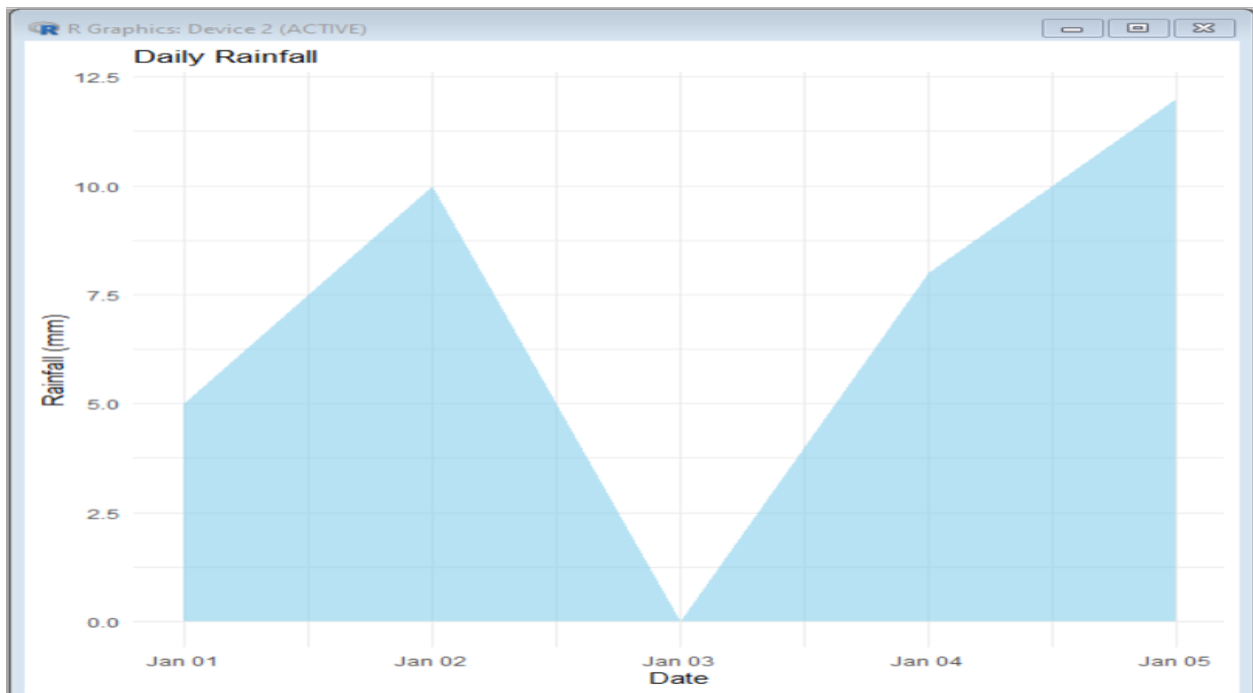
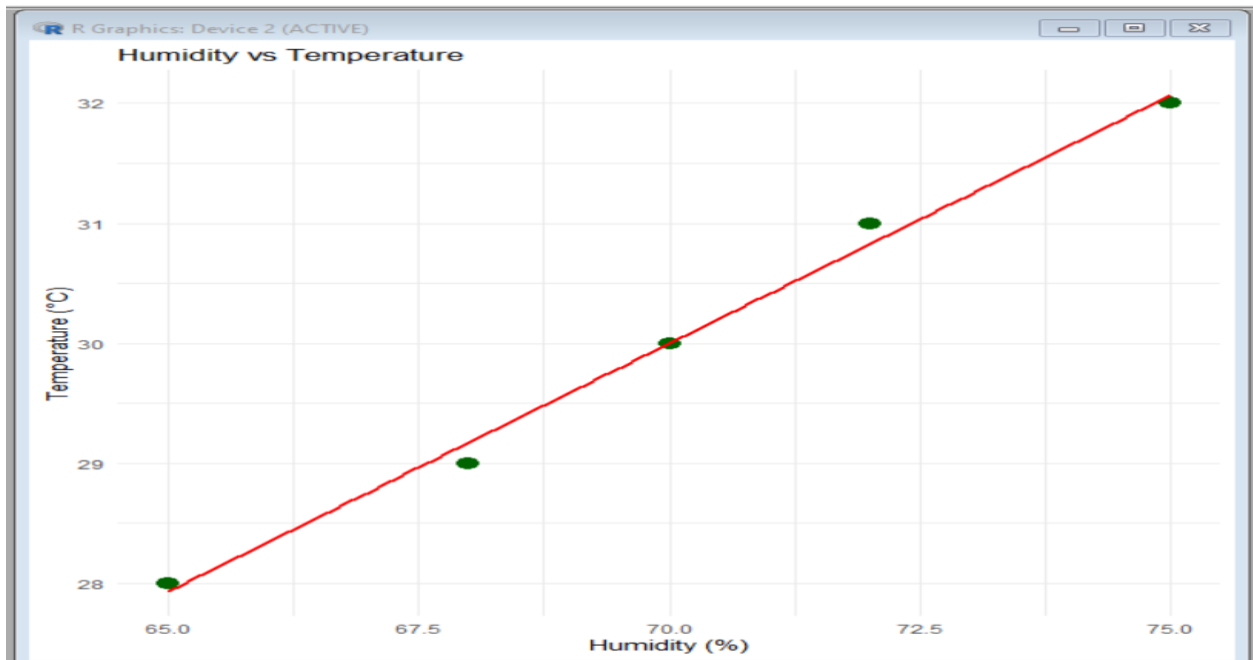
```
p2 <- ggplot(weather_data,  
  aes(x = Humidity, y = Temperature)) +  
  geom_point(color = "darkgreen", size = 3) +  
  labs(title = "Humidity vs Temperature")
```

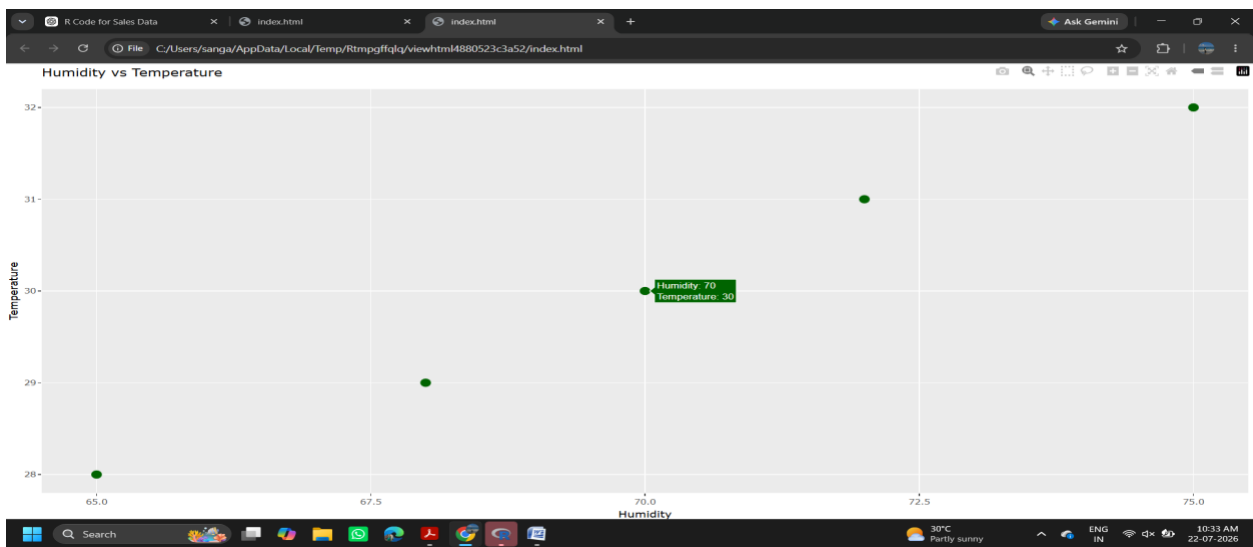
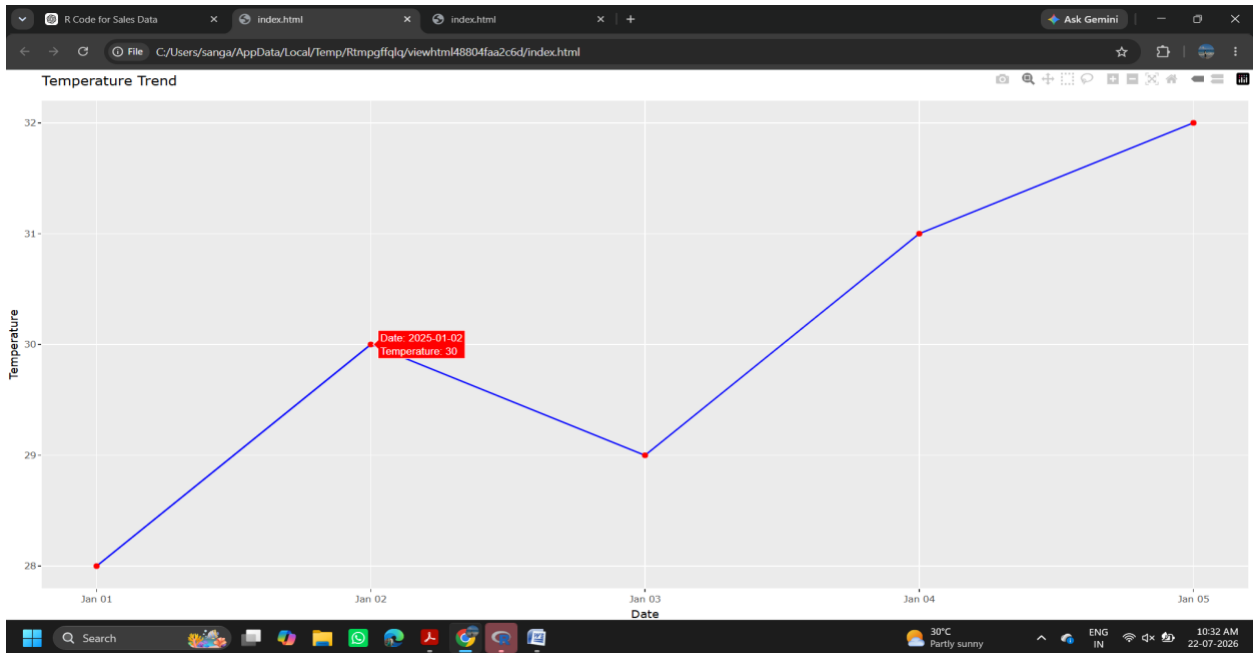
```
ggplotly(p2)
```

```
View(weather_data)
```

Output:







R Data: weather_data

	Date	Temperature	Humidity	Rainfall
1	2025-01-01	28	65	5
2	2025-01-02	30	70	10
3	2025-01-03	29	68	0
4	2025-01-04	31	72	8
5	2025-01-05	32	75	12

Set 12: Dataset: Website Traffic

R code:

```
library(ggplot2)
library(plotly)
```

Dataset

```
health_data <- data.frame(
  Patient_ID = c(1,2,3,4,5),
  Age = c(25,35,45,55,65),
  BMI = c(22.5,26.3,28.1,31.2,29.5),
  Blood_Pressure = c(120,130,135,145,140),
  Cholesterol = c(180,200,220,240,230)
)
```

1. Histogram

```
ggplot(health_data, aes(x = BMI)) +
  geom_histogram(binwidth = 2, fill = "skyblue", color = "black") +
  labs(
    title = "Distribution of BMI",
    x = "BMI",
    y = "Frequency"
  ) +
  theme_minimal()
```

2. Scatter Plot

```
ggplot(health_data,
  aes(x = Age, y = Blood_Pressure)) +
  geom_point(color = "blue", size = 4) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(
    title = "Age vs Blood Pressure",
    x = "Age",
    y = "Blood Pressure"
  ) +
  theme_minimal()
```

3. Line Chart

```
ggplot(health_data,
  aes(x = Age, y = Cholesterol, group = 1)) +
  geom_line(color = "green", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
```

```

title = "Age vs Cholesterol",
x = "Age",
y = "Cholesterol"
) +
theme_minimal()

```

4. Interactive Dashboard

```

p1 <- ggplot(health_data,
             aes(x = Age, y = Blood_Pressure)) +
  geom_point(color = "blue", size = 3) +
  labs(title = "Age vs Blood Pressure")

ggplotly(p1)

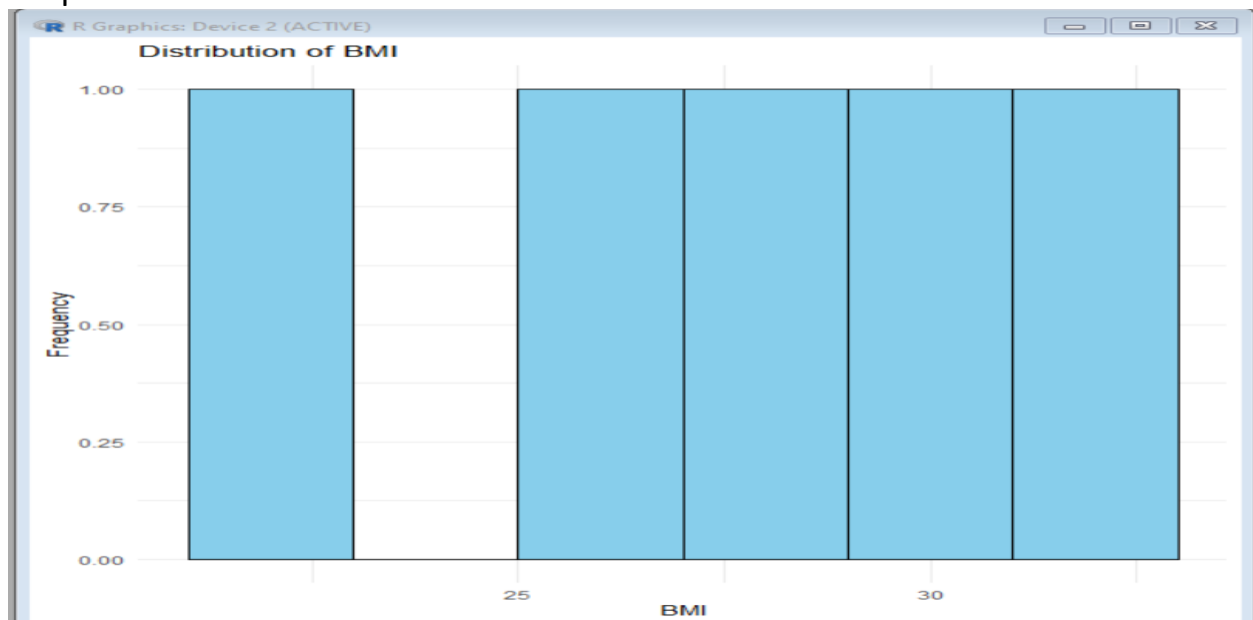
p2 <- ggplot(health_data,
             aes(x = Age, y = Cholesterol, group = 1)) +
  geom_line(color = "green") +
  geom_point(color = "red") +
  labs(title = "Age vs Cholesterol")

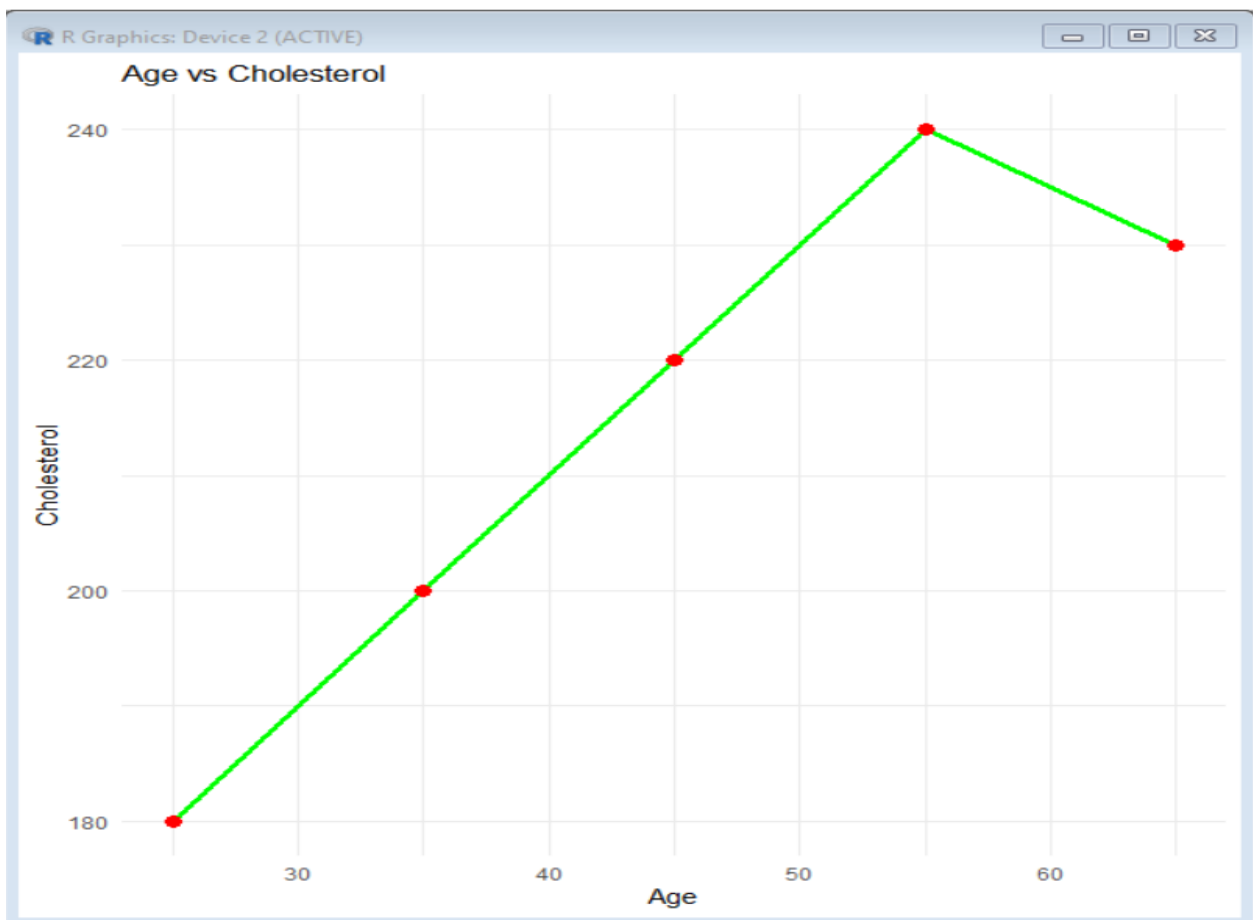
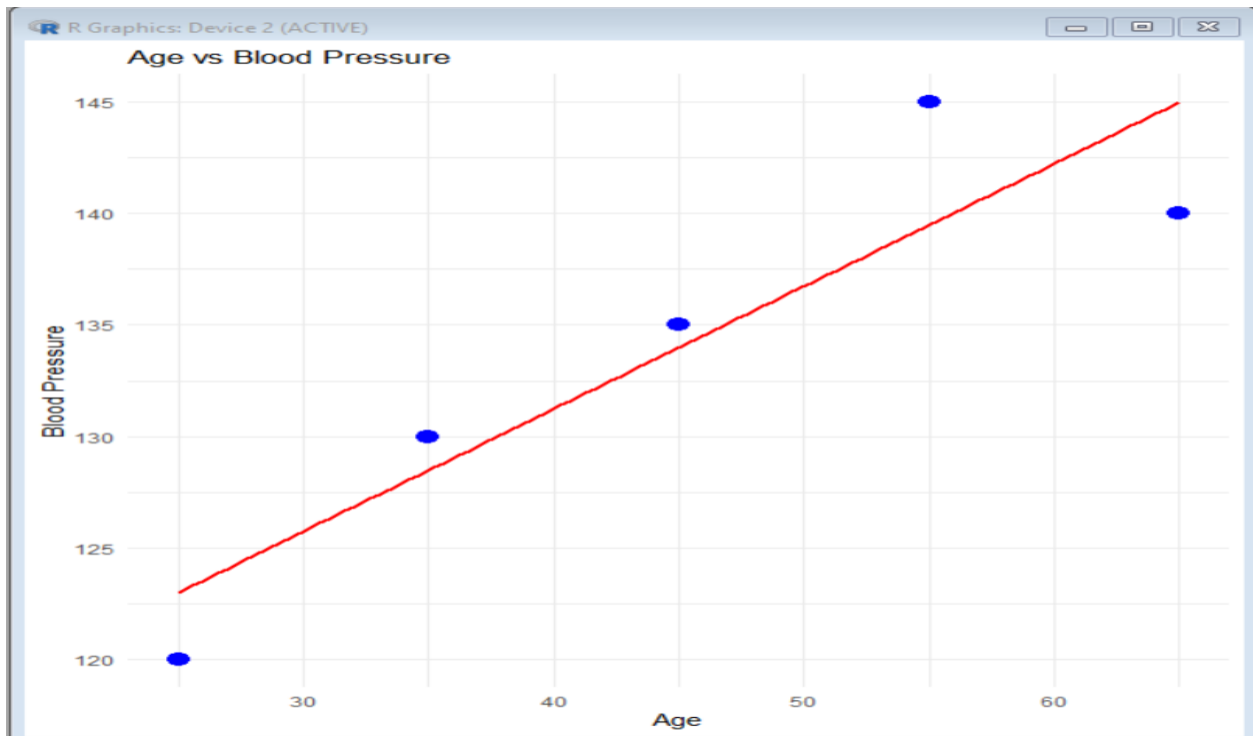
```

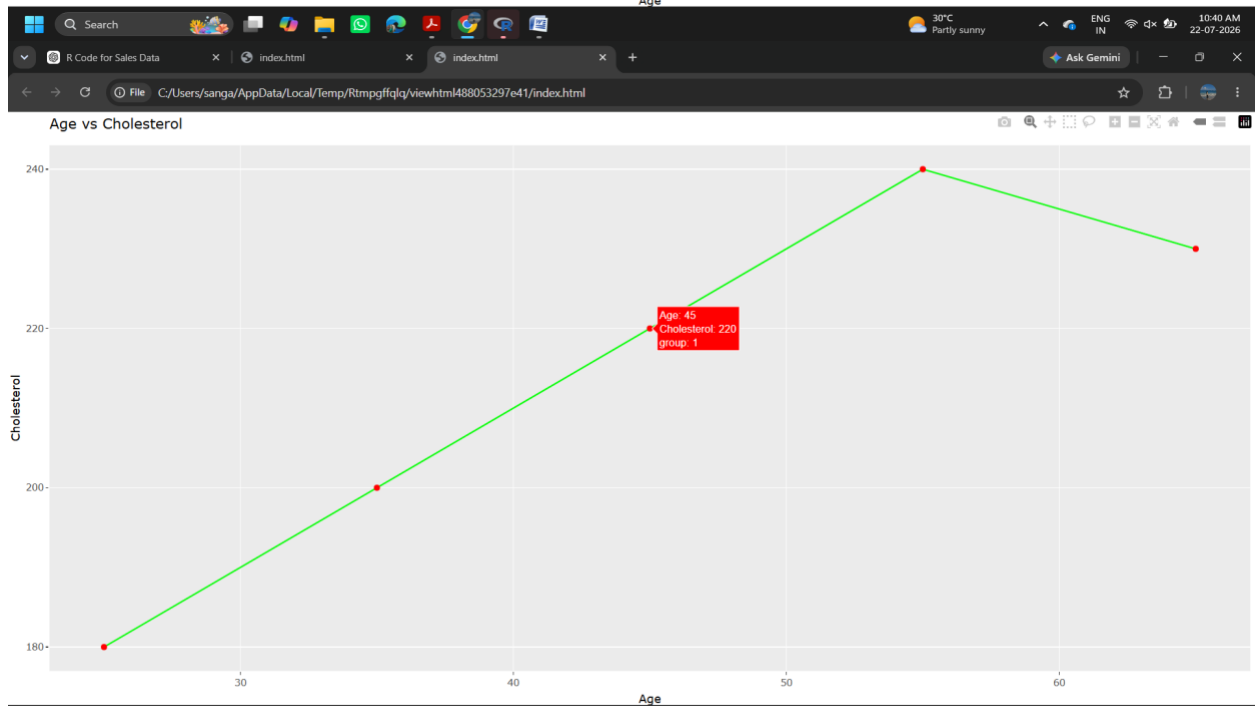
```
ggplotly(p2)
```

```
View(health_data)
```

Output:







R Code for Sales Data | index.html | index.html | Ask Gemini

File C:/Users/sanga/AppData/Local/Temp/Rtmpgffqlq/viewhtml488053297e41/index.html

Data: health_data

	Patient_ID	Age	BMI	Blood_Pressure	Cholesterol
1	1	25	22.5	120	180
2	2	35	26.3	130	200
3	3	45	28.1	135	220
4	4	55	31.2	145	240
5	5	65	29.5	140	230

Set 13. Dataset: Geographic Data

R code:

```
library(ggplot2)
library(plotly)
library(tidyr)

# Dataset
energy_data <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Solar = c(120, 140, 160, 180, 200),
  Wind = c(100, 110, 130, 150, 170),
  Hydro = c(80, 85, 90, 95, 100)
)
```

1. Grouped Bar Chart

```
energy_long <- pivot_longer(
  energy_data,
  cols = c(Solar, Wind, Hydro),
  names_to = "Source",
  values_to = "Production"
)

ggplot(energy_long,
  aes(x = Month,
      y = Production,
      fill = Source)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Monthly Renewable Energy Production",
    x = "Month",
    y = "Energy Production"
  ) +
  theme_minimal()
```

2. Stacked Area Chart

```
ggplot(energy_long,
  aes(x = Month,
      y = Production,
      fill = Source,
      group = Source)) +
  geom_area() +
  labs(
    title = "Renewable Energy Production Trend",
    x = "Month",
    y = "Energy Production"
  ) +
```

```
theme_minimal()
```

```
# 3. Table  
print(energy_data)
```

```
# 4. Interactive Dashboard
```

```
p1 <- ggplot(energy_long,  
  aes(x = Month,  
      y = Production,  
      fill = Source)) +  
  geom_bar(stat = "identity", position = "dodge") +  
  labs(title = "Monthly Energy Production")
```

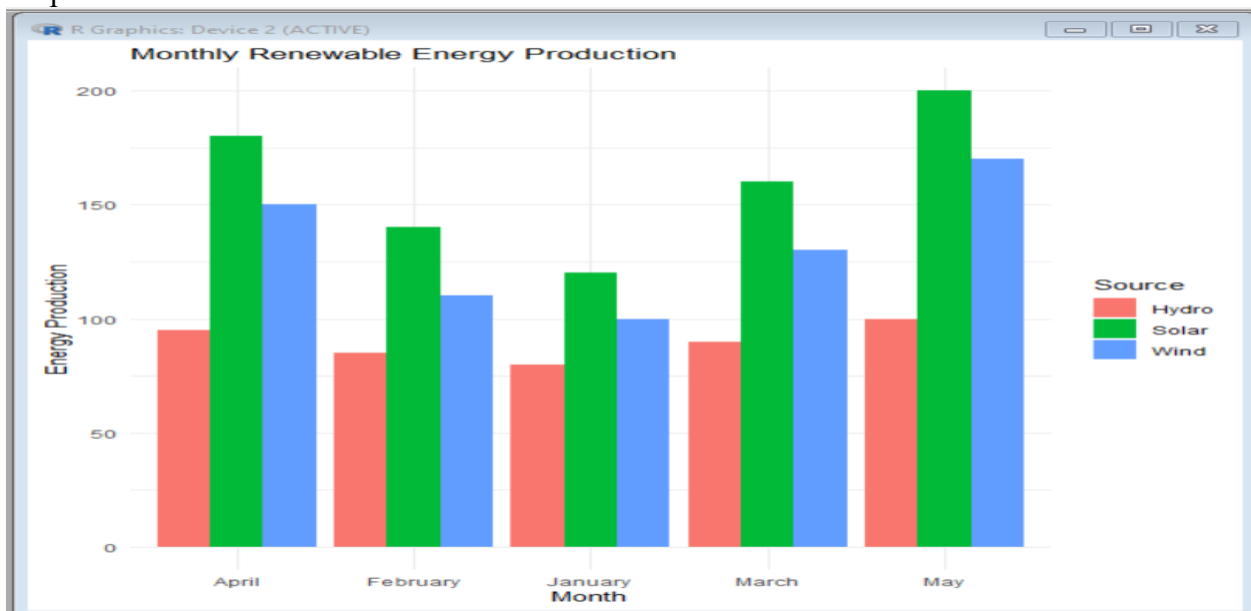
```
ggplotly(p1)
```

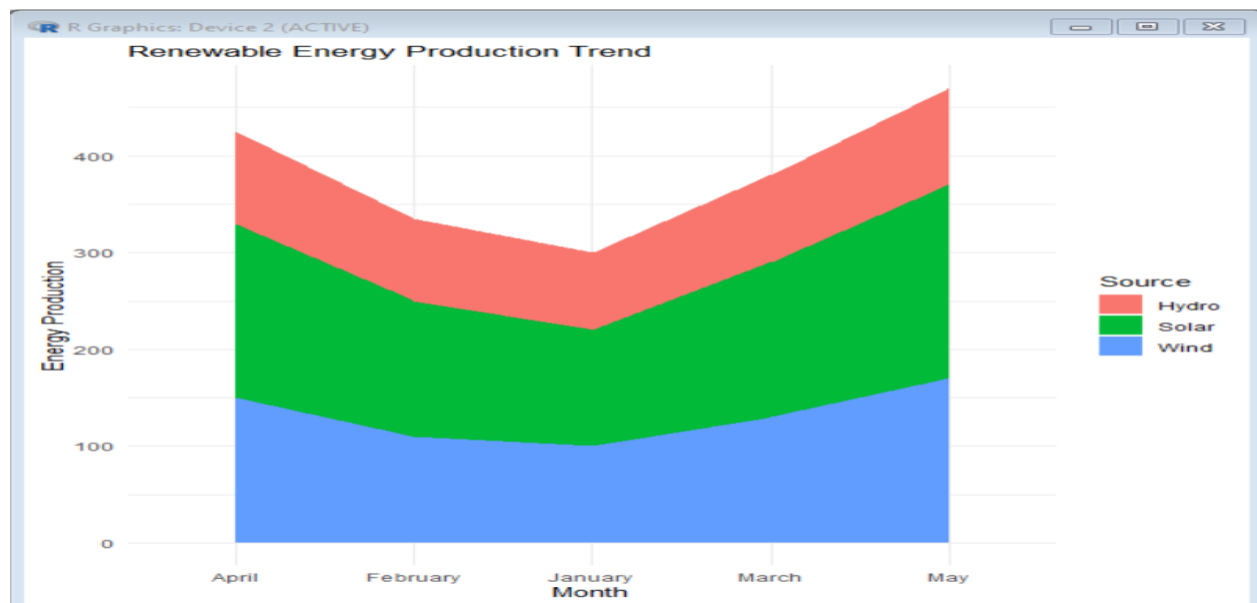
```
p2 <- ggplot(energy_long,  
  aes(x = Month,  
      y = Production,  
      fill = Source,  
      group = Source)) +  
  geom_area() +  
  labs(title = "Energy Production Trend")
```

```
ggplotly(p2)
```

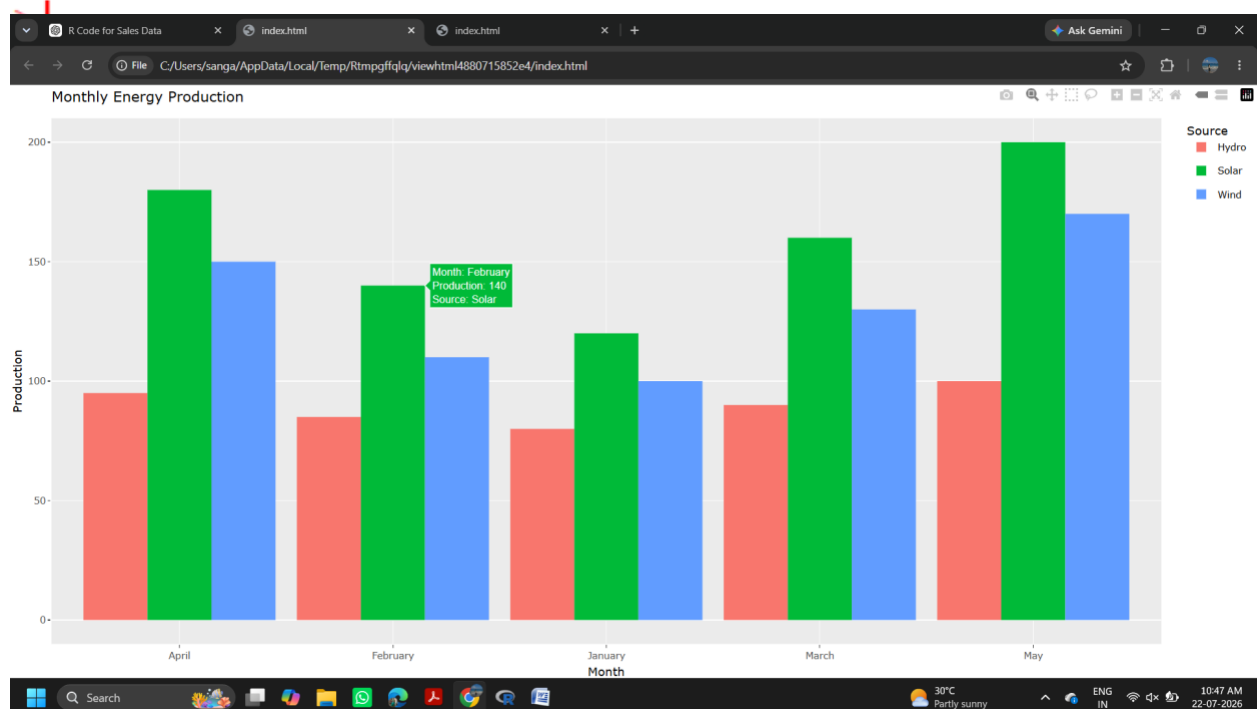
```
View(energy_data)
```

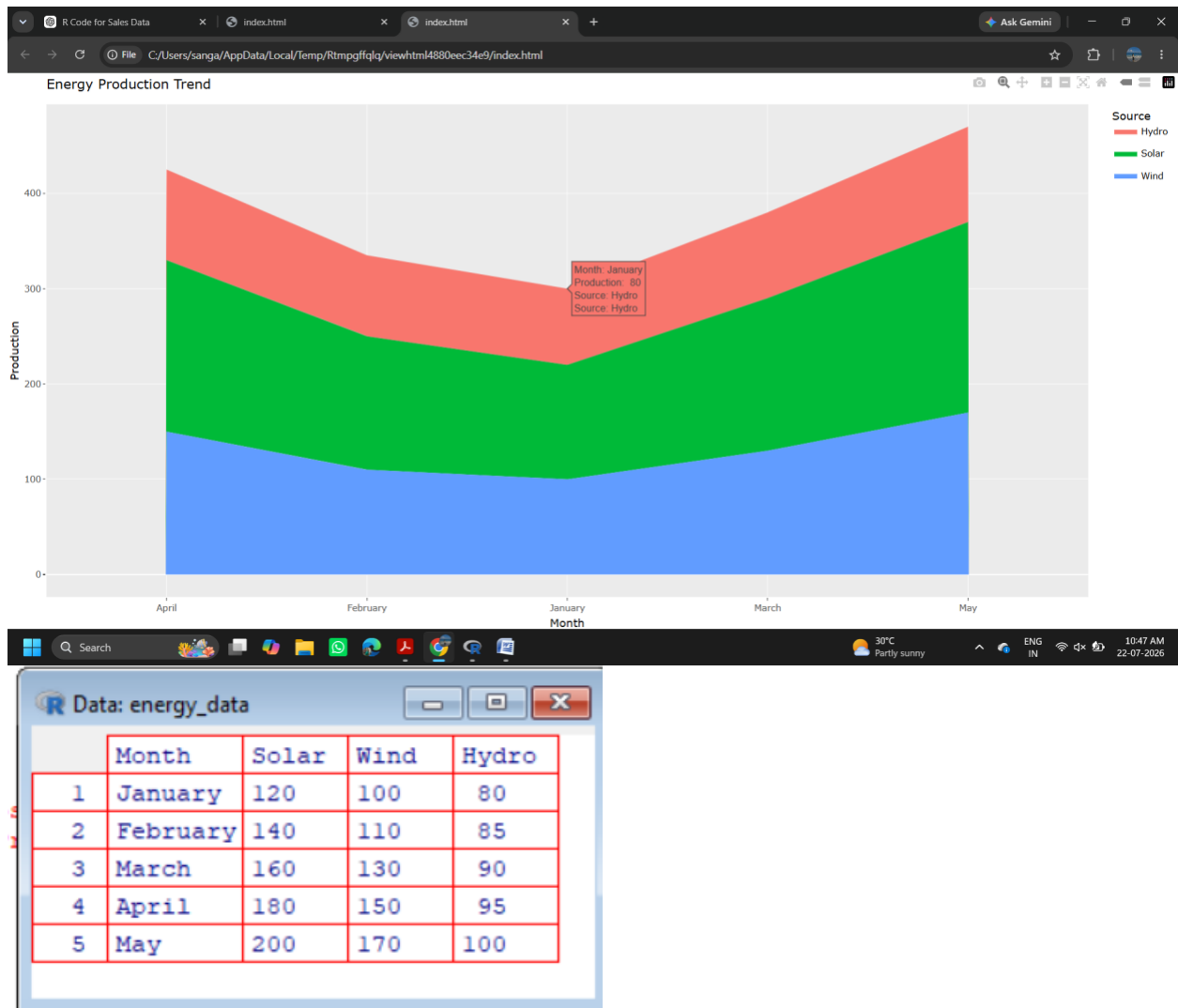
Output:





```
Month Solar Wind Hydro
1 January 120 100 80
2 February 140 110 85
3 March 160 130 90
4 April 180 150 95
5 May 200 170 100
>
```





Set 14. Dataset: Survey Results

R code:

```
library(ggplot2)
```

```
library(plotly)
```

```
library(tidyr)
```

```
# Dataset
```

```
traffic_data <- data.frame(
  Time = c("8 AM", "9 AM", "10 AM", "11 AM", "12 PM"),
  Cars = c(120, 180, 150, 170, 160),
  Bikes = c(40, 55, 50, 60, 58),
  Buses = c(20, 25, 22, 28, 24)
)
```

```
# 1. Line Chart
```

```

traffic_long <- pivot_longer(
  traffic_data,
  cols = c(Cars, Bikes, Buses),
  names_to = "Vehicle",
  values_to = "Count"
)

ggplot(traffic_long,
  aes(x = Time,
    y = Count,
    color = Vehicle,
    group = Vehicle)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
  labs(
    title = "Traffic Volume by Time",
    x = "Time",
    y = "Vehicle Count"
  ) +
  theme_minimal()

```

2. Stacked Bar Chart

```

ggplot(traffic_long,
  aes(x = Time,
    y = Count,
    fill = Vehicle)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Vehicle Distribution by Time",
    x = "Time",
    y = "Vehicle Count"
  ) +
  theme_minimal()

```

3. Table

```
print(traffic_data)
```

4. Interactive Dashboard

```

p1 <- ggplot(traffic_long,
  aes(x = Time,
    y = Count,
    color = Vehicle,
    group = Vehicle)) +

```

```
geom_line() +  
geom_point() +  
labs(title = "Traffic Volume")
```

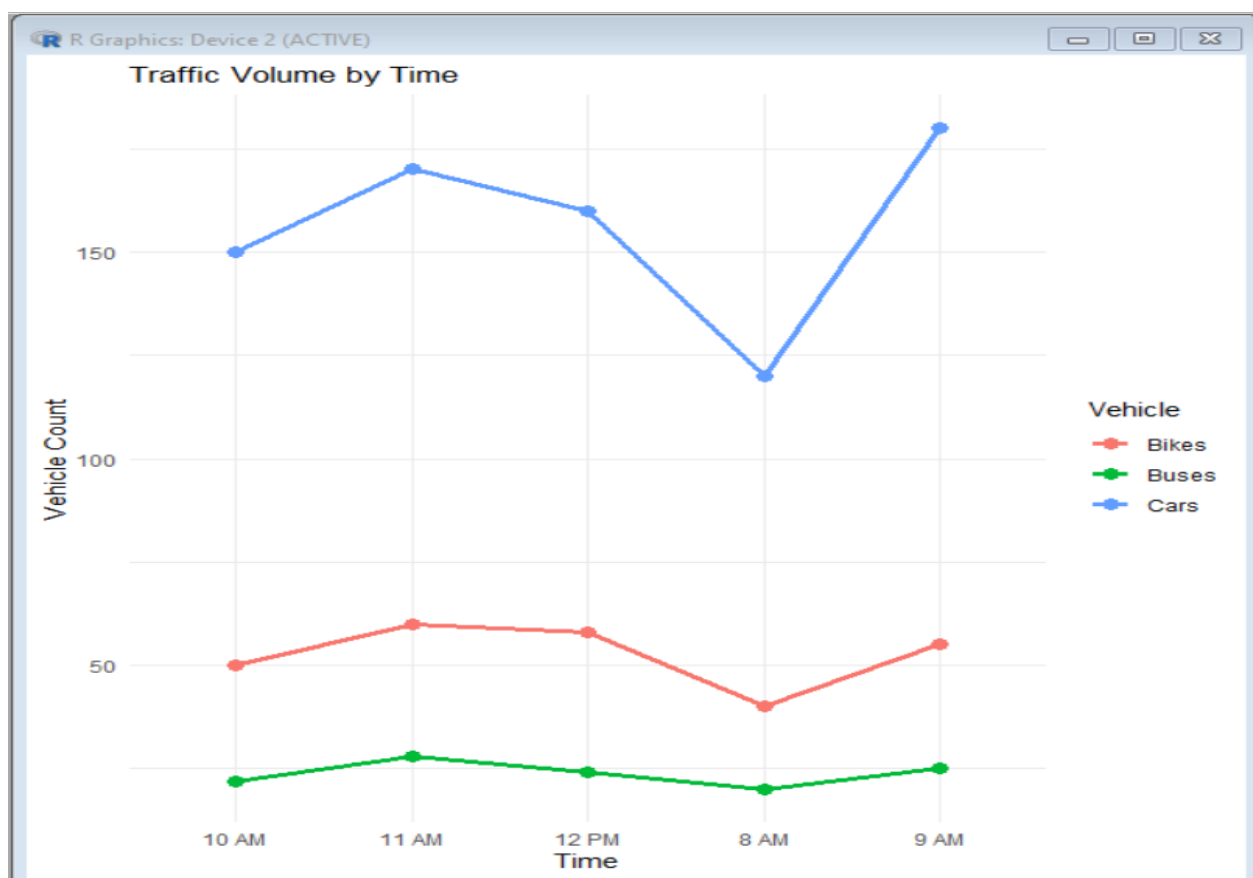
```
ggplotly(p1)
```

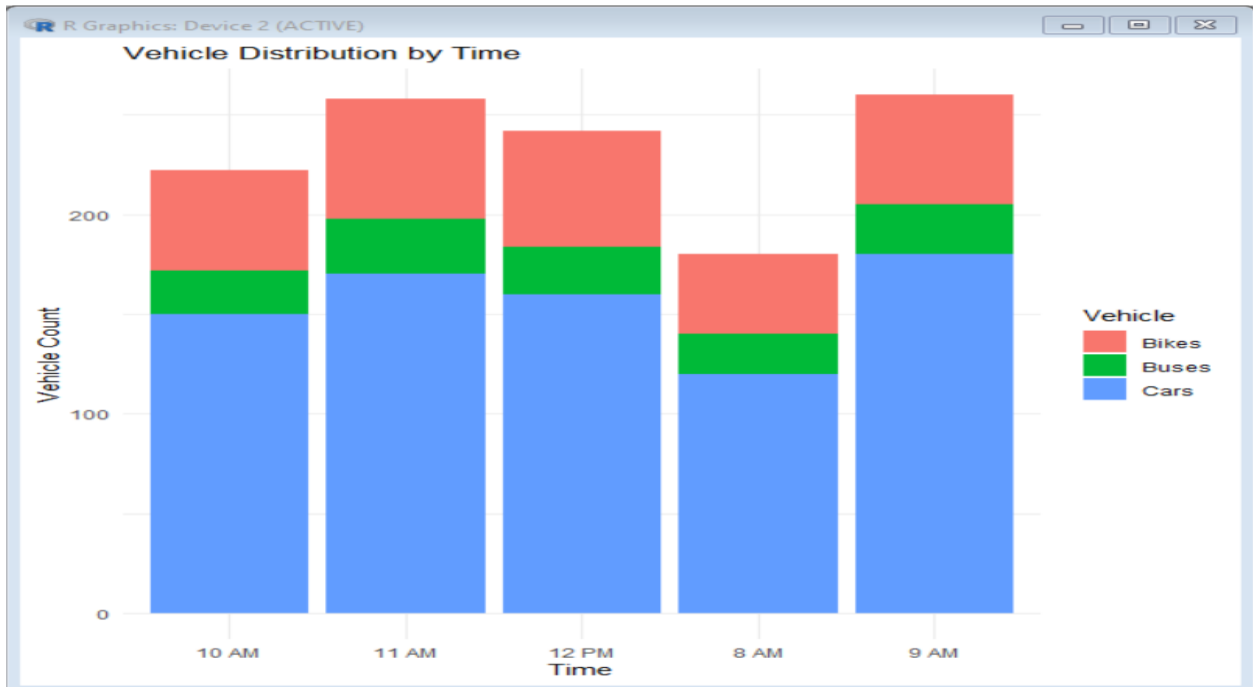
```
p2 <- ggplot(traffic_long,  
  aes(x = Time,  
    y = Count,  
    fill = Vehicle)) +  
  geom_bar(stat = "identity") +  
  labs(title = "Vehicle Distribution")
```

```
ggplotly(p2)
```

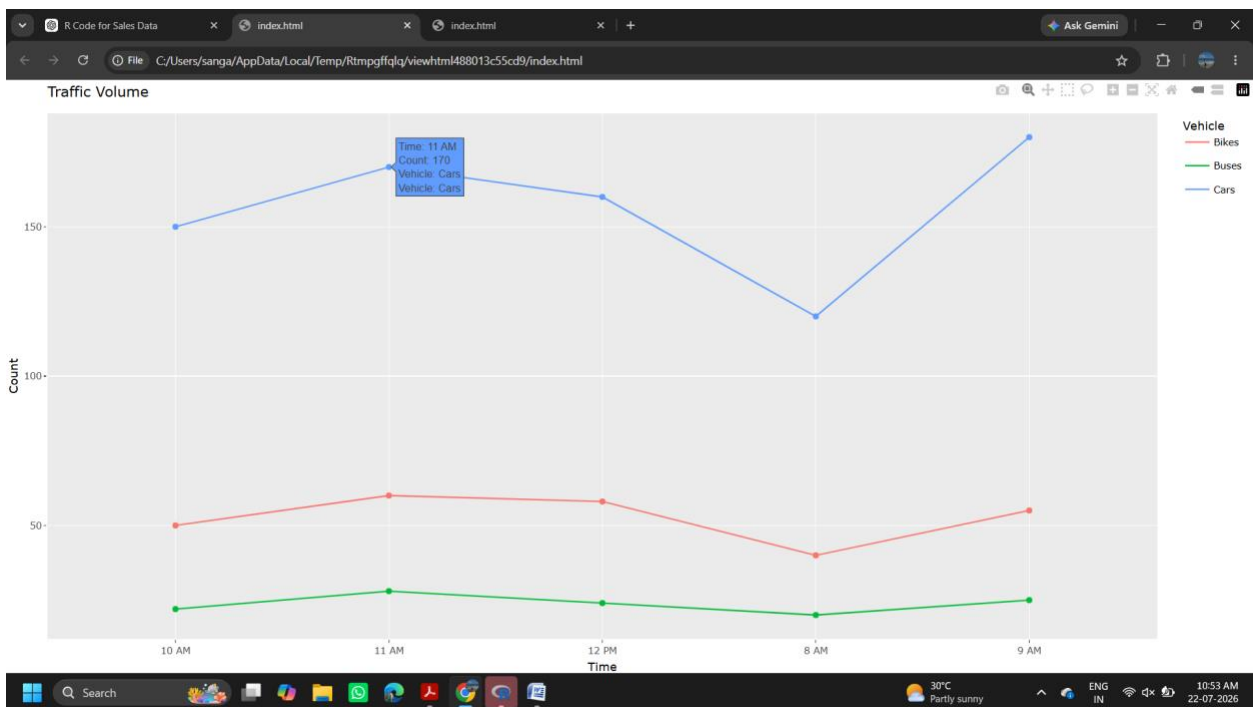
```
View(traffic_data)
```

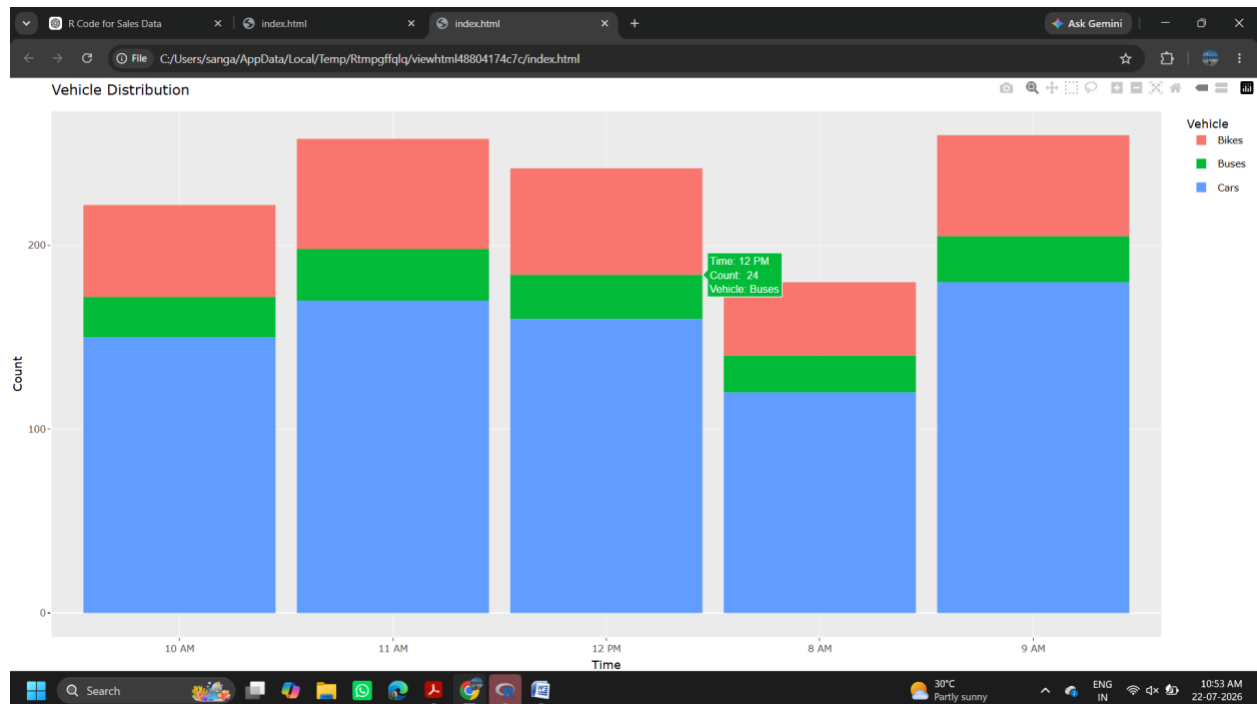
Output:





```
Time Cars Bikes Buses
1 8 AM 120 40 20
2 9 AM 180 55 25
3 10 AM 150 50 22
4 11 AM 170 60 28
5 12 PM 160 58 24
```





	Time	Cars	Bikes	Buses
1	8 AM	120	40	20
2	9 AM	180	55	25
3	10 AM	150	50	22
4	11 AM	170	60	28
5	12 PM	160	58	24

Set 15: Product Sales Analysis

R code:

```
library(ggplot2)
```

```
library(plotly)
```

```
library(tidyr)
```

```
# Dataset
```

```
movie_data <- data.frame(
```

```
  Movie = c("Movie A", "Movie B", "Movie C", "Movie D", "Movie E"),
```

```
  Action = c(80, 70, 90, 75, 85),
```

```
  Comedy = c(60, 75, 65, 80, 70),
```



```
Drama = c(90,85,88,92,87)
)
```

```
# Convert to Long Format
movie_long <- pivot_longer(
  movie_data,
  cols = c(Action, Comedy, Drama),
  names_to = "Genre",
  values_to = "Rating"
)
```

```
# 1. Grouped Bar Chart
```

```
ggplot(movie_long,
  aes(x = Movie,
      y = Rating,
      fill = Genre)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Movie Ratings by Genre",
    x = "Movie",
    y = "Rating"
  ) +
  theme_minimal()
```

```
# 2. Heatmap
```

```
ggplot(movie_long,
  aes(x = Genre,
      y = Movie,
      fill = Rating)) +
  geom_tile(color = "white") +
  labs(
    title = "Movie Ratings Heatmap",
    x = "Genre",
    y = "Movie"
  ) +
  theme_minimal()
```

```
# 3. Table
```

```
print(movie_data)
```

```
# 4. Interactive Dashboard
```

```
p1 <- ggplot(movie_long,
  aes(x = Movie,
```

```
    y = Rating,  
    fill = Genre)) +  
  geom_bar(stat = "identity", position = "dodge") +  
  labs(title = "Movie Ratings")
```

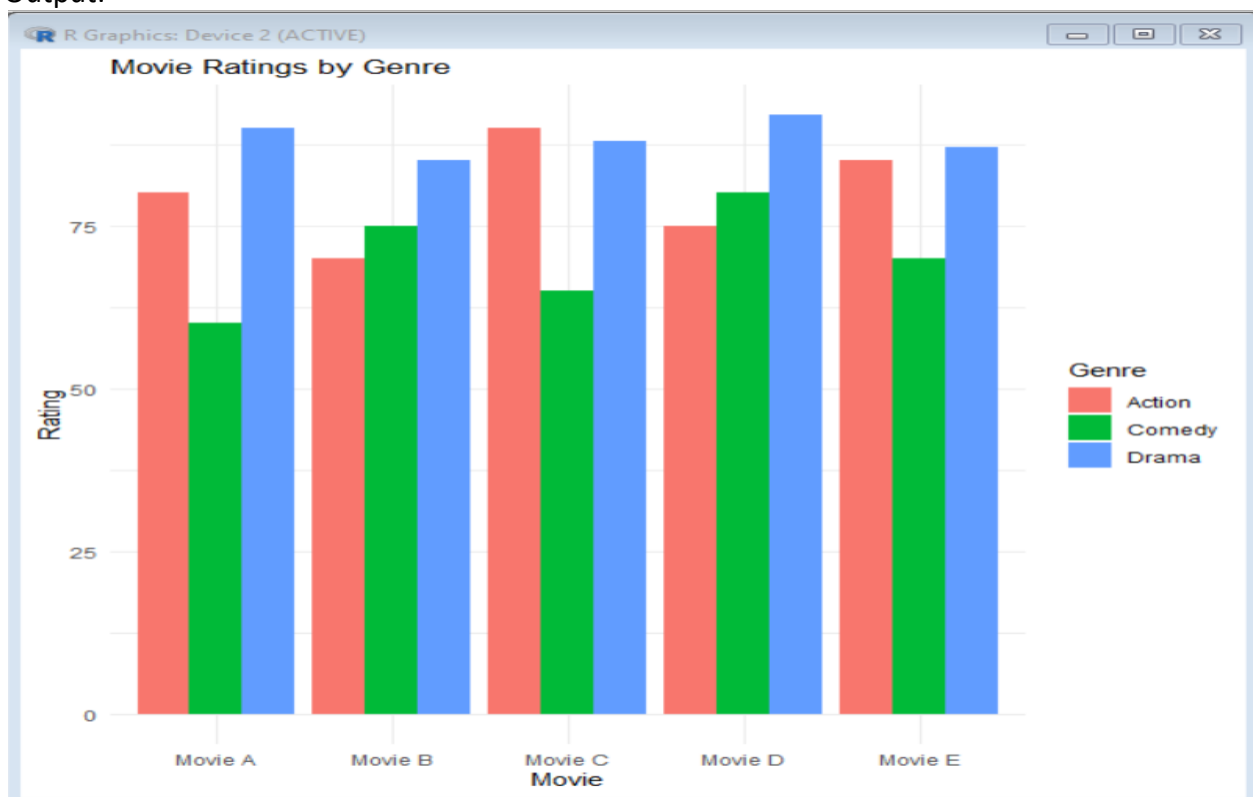
```
ggplotly(p1)
```

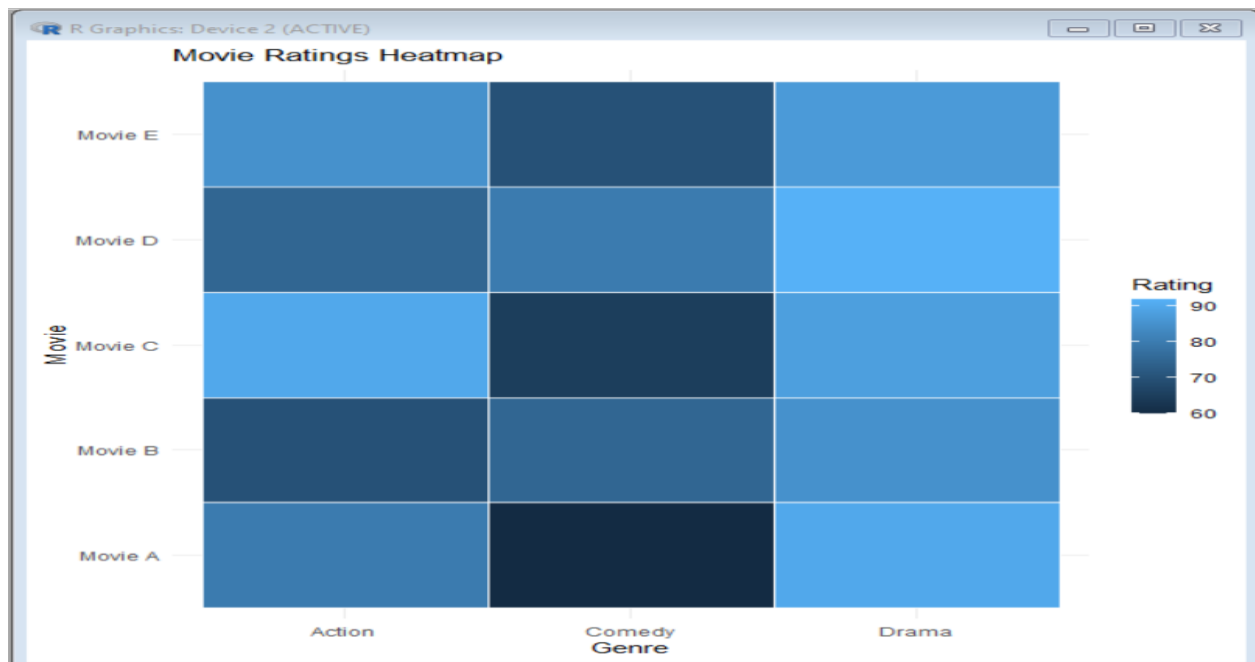
```
p2 <- ggplot(movie_long,  
  aes(x = Genre,  
      y = Movie,  
      fill = Rating)) +  
  geom_tile() +  
  labs(title = "Movie Ratings Heatmap")
```

```
ggplotly(p2)
```

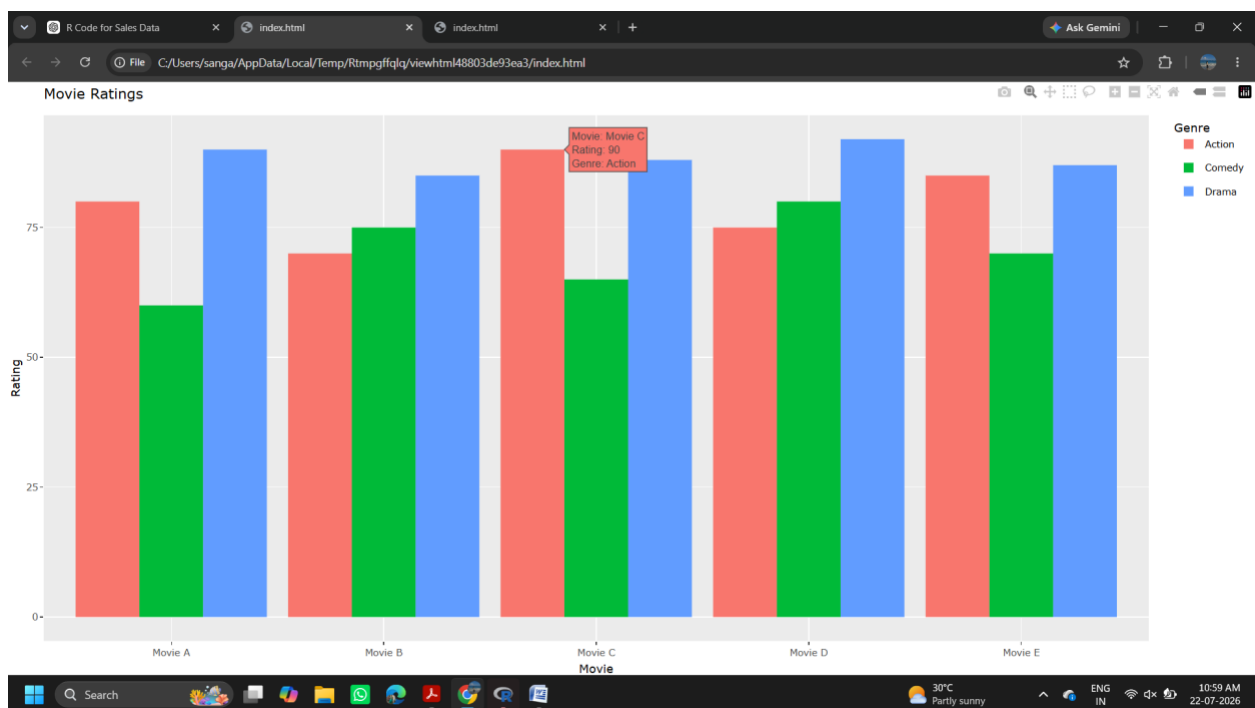
```
View(movie_data)
```

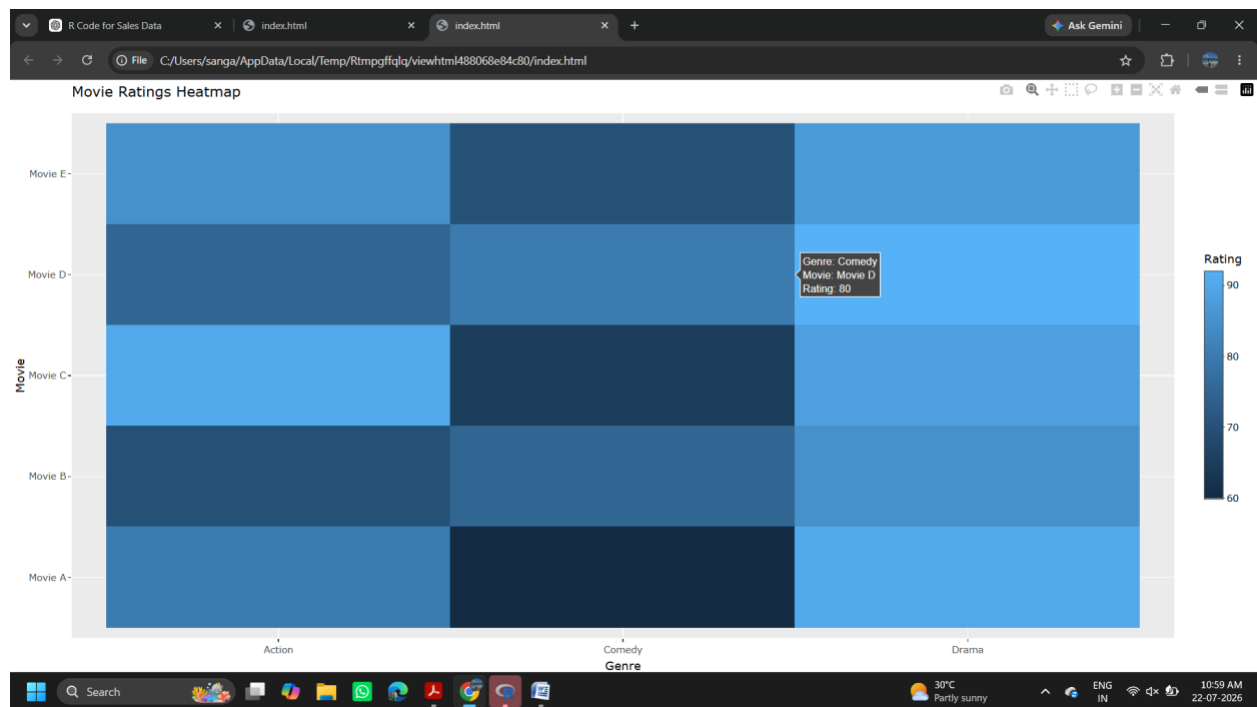
Output:





```
> print(movie_data)
  Movie Action Comedy Drama
1 Movie A      80      60      90
2 Movie B      70      75      85
3 Movie C      90      65      88
4 Movie D      75      80      92
5 Movie E      85      70      87
>
```





R Data: movie_data

	Movie	Action	Comedy	Drama
1	Movie A	80	60	90
2	Movie B	70	75	85
3	Movie C	90	65	88
4	Movie D	75	80	92
5	Movie E	85	70	87